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(1) Research Program/Project/Study Title: DEVELOPMENT OF TROLLEY TYPE DRAWING AND ART TABLE	
Proponent Name:	Warren S. Celoso
Academic Rank:	Associate Professor III
Co-Authors (with academic rank):	Prof. Eduardo F. Famucol (Asso. Prof. IV)
Campus/Satellite College/Address:	CapSU Main Campus
Telephone/Cell Phone/Fax Number:	09124759683
E-mail Address:	warzceloso06@gmail.com

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Roxas City
Province:	Capiz
Region:	VI

(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☒ Applied	☐ Tech. Promotion/Commercialization						

(7) Sector/Commodity Furniture Business DTI (Department of Trade and Industry) DepEd- TLE	(8) Discipline/Field of Specialization Drafting Architecture Civil Engineering
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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The researchers develop trolley type drawing and art table for research study with the desire to improve, enhance usefulness, mobility and practical ways for drawing and painting activities especially on school based art making project, commissioned artworks as well as job satisfaction and for economic development.

In this problem, a study that somehow can be used as an alternative for conventional single component painting stand and drawing table which are available in the market. To make research, do some experiments, share ideas, knowledge, and thoughts in able to develop trolley type drawing table with painting stand as an stand alone single unit for drawing and painting: By this information stated, it can help and improve the skills of each individual. This study enhances the performance level of every person who wants to use this product.

A kind of device can be use to provide ease of work in terms of drawing and painting activities. The essence of this research is to add feature of the device and mobility for workmanship, it will enhance draftsman and artist capability to work using this device. Perhaps it is difficult as first time to introduce this kind of equipment but to maintain the ability of innovative mind when it comes to technical specification, performance and applicability surely we can able to adapt and develop trolley type drawing table with painting stand. Objectives of the Study: Generally, this study attempted to develop trolley type drawing table with painting stand. Specifically, it aimed to; (1) evaluate the trolley type drawing and art table in terms of utility, composition and portability.(2) applicability of trolley type drawing and art table in terms of design, capacity and efficiency.(3) acceptability of trolley type drawing and art table as to its technical specification and safety features.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Composition, Utility, Portability, technical specification and feature

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

The design is actually the product of continuous effort of the researchers to come up with a device suitable for mobility of making engineering drawing and art activities.

In designing, the researcher will be developing trolley type drawing and art table with wooden type top coated paint, components and mechanical parts which is properly joint and mechanically fastened with their specific functions. The design is composed of three main parts which utilized the drawing board, art painting stand and drawing/painting accessories which are interconnected to each other whenever the device is used. It also include body frame and ball caster wheel whenever the device is used. The device can be operational and functional according to each parts and it will be done through manual operation.

During the Industrial Revolution draftsman ship gradually became a specialized trade and drawing tables slowly moved out of the libraries and offices of most gentlemen. They became more utilitarian and were built of steel and plastic instead of fine woods and brass (Baskinger and Bardel, 2013).

Very often a drawing table could look like a writing table or even a pedestal desk when the working surface was set at the horizontal and the height adjusted in order to use it as a "normal" desk. The only giveaway was usually a lip on one of the sides of the desktop. This lip or edge stopped paper or books from sliding when the surface was given an angle. It was also sometimes used to hold writing implements. When the working surface was extended at its full height, a drawing table could be used as a standing desk (Gillis and Hammer, 2017).

Easels became critically important as artists began to explore their natural world. Easily portable, easels enabled artists to paint directly from nature, opening up a whole new world of light and colour based on actual, real-time observation. Arguably, this form of plain air expression gained its greatest momentum in the 19th century with the Barbizon school and the growing interest in impressionism.

And of course today, the easel remains an important piece of equipment for artists of all skill levels. In fact, we can see easels being used as early as high school to enable student to develop fine motor skills and express themselves, all in the fun guise of creating art (Madison, 2002).

Recently, no available modified drafting table with chair and easel stand in the market which is multi-purpose. Thus, the researcher was motivated to innovate a table and easel which are different from conventional drafting table and painting stand. An easel makes your work easy and comfortable ergonomically and help create accurate painting. It also keep perspective precise upon using the said device. Like many other essential materials for a good piece of art, an easel too is an undeniably important part of any artist's gear.

Likewise, some experts might influence the purpose of developing and innovating the process of drawing on a table from the traditional way of setting up a drawing into drawing table. There are still other means of setting up a drawing table that has a light functionality for draftsman to used. People are engaged to produce other way to minimize effort in dealing with a dim room or a place to have a lit drawing table.

Objectives of the Study

Generally, this study attempted to develop trolley type drawing and art table. Specifically, it aimed to; (1) evaluate the trolley type drawing and art table in terms of utility, composition and portability.

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- (2) applicability of trolley type drawing and art table in terms of design, capacity and efficiency.
(3) acceptability of trolley type drawing and art table as to its technical specification and safety features.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The materials used for the development of trolley type drawing and art table were fabricated with the following; Marine type plyboard 1/2" x 18" x 24" are utilized, water resistant and compatible for indoor & outdoor art activities. 1 3/4" x 12" x 10" Drawer type fabricated as storage area for drawing instruments. 1/2" x 24" length attached metal plate hinges are fastened tightly for the used of 30 degree inclined position of the drawing table as painting stand. 1" x 2" x 20" integrated with foldable legs and 1" x 2" x 18" bracket system for assembling in upright position with bolt and nut connected to a wooden legs as extension for balance and support of the table. Attached also are 2" diameter ball caster wheel which are foldable for mobility and ease of transportation. While, layout and plan for fabrication are being followed, construction of the device are connected properly and packaging are well-covered or decorated with paint.

This study is the developmental method of research, developmental method has been defined as "the systematic study of designing, developing and evaluating instructional programs, processes, and products that must meet the criteria of internal consistency and effectiveness" (Seels & Richey, 1994). The development of trolley type drawing and art table is the modification of the different processes involved in working with mechanical drawing and artwork production in painting.

Construction Procedure	Number of months								
	1	2	3	4	5	6	7	8	9
Designing /layout of the device and electrical components and installation	■								
Pre selection of materials		■							
Fabrication			■						
Installation/ Assemble of Components				■					
Checking and review of structure				■	■				
Construction of the different parts						■			
Product Testing						■	■		
Testing Revision							■		
Observation and Data Recording								■	
Evaluation									■

Location of the study was done in the Drafting technology Laboratory Room of the Industrial Technology department, CapSU Main Campus.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Table 1. Evaluate the qualitative description of develop trolley type drawing and art table in terms of;

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Utility	Mean	Verbal Interpretation
1. Planned for a specific purpose in technical drawing and visual art activities..	4.50	Very Usable
2. Parts can be assemble & dis assemble for workmanship.	4.30	Very Usable
3. Its function are applicable for end user having basic and technical skills in drawing and painting.	4.35	Very Usable
Grand Mean	4.38	Very Usable
Composition		
1. Made of 1/2 x 18 x 24" marine ply board and wooden framing coated with epoxy primer paint and properly fastened its parts and joints.	4.42	Very Good
2. Hinges are also attached purposely in terms of inclined/slanting position for art or painting activities.	4.35	Very Good
3. 1" x 2" x 30" Foldable/removable legs with adjustable bolt & nut and bracing are made of wooden stick for conversion of trolley type drawing table	4.30	Very Good
Grand Mean	4.36	Very Good
Portability		
1. Ability to move or be moved freely and easily.	4.50	Very Portable
2. Ability to fold and convert drawing board into trolley type device.	4.25	Very Portable
3. Ball-caster wheel are attach for mobility.	4.45	Very Portable
Grand Mean	4.40	Very Portable

The data showed in **table 1**, that the qualitative description of the develop trolley type drawing and art table in terms of **utility**, a grand mean of **4.38** which verbally interpreted as **very usable**. This implies that the device planned for a specific purpose in technical drawing and visual art activities, parts can be assemble & dis assemble for workmanship and its function are applicable for end user having basic and technical skills in drawing and painting. In terms of **composition**, a grand mean of **4.36** which verbally interpreted as **very good** which implies that the parts are made of 1/2 x 18 x 24" marine ply board and wooden framing coated with epoxy primer paint and properly fastened its parts and joints, hinges are also attached purposely in terms of inclined/slanting position for art or painting activities with 1" x 2" x 30" foldable/removable legs with adjustable bolt & nut and bracing are made of wooden stick for conversion of trolley type drawing table.

In terms of **portability**, a grand mean of **4.40** which verbally interpreted as **very portable** which implies that the device have the ability to move or be moved freely and easily, also ability to fold and convert drawing board into trolley type device with ball-caster wheel are attach for mobility.

Table 2. Applicability of trolley type drawing and art table in terms of design, capacity and efficiency.

Design	Mean	Verbal Interpretation
1. Layout and plan for fabrication are being followed.	4.45	Very Applicable
2. Tools and materials of the device are being used for construction.	4.35	Very Applicable
3. Packaging are well-covered or decorated with paint.	4.50	Very Applicable
Grand Mean	4.43	Very Applicable
Capacity		
1. A total measurement of 4" depth, 18" in width and 24" in length drawing board with drawer type weighing 1.5 kilos.	4.40	Very Applicable
2. A wooden leg and bracing are integrated in converting trolley into drawing table measuring 1" thickness x 2" width x 30" in height with drawer and ball-caster	4.45	Very Applicable

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attached. 3. A total load of 15 kilos are applied and constructed for the fabrication of the device.	4.35	Very Applicable
Grand Mean	4.40	Very Applicable
Efficiency		
1. Minimum of 5 to 10 minutes assembling and dis assembling with proper orientation.	4.30	Very Applicable
2. Capable to used 8 hours a day for mechanical drawing and art activities.	4.36	Very Applicable
3. Effective used for secondary and college students in terms of mechanical drawing and visual art activities.	4.32	Very Applicable
Grand Mean	4.32	Very Applicable

The data showed in **table 2**, that the Applicability of develop trolley type drawing and art table in terms of **design**, a grand mean of **4.43** which verbally interpreted as **very applicable**. This implies that the device layout and plan for fabrication are being followed, tools and materials of the device are being used for construction and packaging are well-covered or decorated with paint. In terms of **capacity**, a grand mean of **4.40** which verbally interpreted as **very applicable** which implies that the device a total measurement of 4" depth, 18" in width and 24" in length drawing board with drawer type weighing 1.5 kilos, a wooden leg and bracing are integrated in converting trolley into drawing table measuring 1" thickness x 2" width x 30" in height with drawer and ball-caster attached with a total load of 15 kilos are applied and constructed for the fabrication of the device. In terms of **efficiency**, a grand mean of **4.32** which verbally interpreted as **very applicable** which implies that the device with a minimum of 10 minutes assembling and dis assembling with proper orientation, capable to used 8 hours a day for mechanical drawing and art activities and effective used for secondary and college students in terms of mechanical drawing and visual art activities.

Table 3. Acceptability of trolley type drawing and art table as to its technical specification and features of the device.

Technical specification and feature of the device	Mean	Verbal Interpretation
1.)4" depth, 18" in width and 24" in length drawing board made of marine type ply board measuring coated with lacquer spray paint and water resistant	4.50	Very Acceptable
2) A dimension of 1" x 2" x 30" wooden leg and support bracing are attached in converting trolley type drawing table integrated with drawer type table.	4.30	Very Acceptable
3) 1/2 x 1/2 x 20" piano hinges are attached for inclined/slanting position of the drawing table into painting stand.	4.65	Very Acceptable
4) 1/2 thick x 3/4" wide x 10" length adjustable wooden bracket connected to drawing board in inclined/slanting position for art activities.	4.40	Very Acceptable
5) Attached also are foldable and removable wooden bracing with the dimension of 1" thickness by 2" in	4.30	Very Acceptable

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width and 18" in length with ball caster wheel attach for mobility and ease of transportation.

Grand Mean

4.43

Very Acceptable

The data showed in **table 3**, that the the acceptability of develop trolley type drawing and art table stand considering its technical specification and feature of the device got a grand mean of **4.43** which verbally interpreted as **very acceptable**. This implies that attached foldable and removable wooden bracing with the dimension of 1" thickness by 2" in width and 18" in length with ball caster wheel attach for mobility and ease of transportation got the lowest mean of **4.30** which verbally interpreted also as **very acceptable**. While, 1/2 x 1/2 x 20" piano hinges are attached for inclined/slanting position of the drawing table into painting stand got a highest mean of **4.65** verbally interpreted as **very acceptable**.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the findings of the study, the following conclusions were drawn:

The qualitative description of develop trolley type drawing and art table in terms of utility, composition and portability enhance ease of work with adjustable mechanism for painting stand and workmanship for technical/mechanical drawing.

Moreover, the Applicability of develop trolley type drawing and art table in terms of design, capacity and efficiency by providing the layout and planned accordingly with its parts and components for fabrication or construction as well as to its mobility upon working in an indoor/outdoor drawing and art activities which will benefit the consumer or end user.

While acceptability of trolley type drawing table and painting stand in terms of the different technical specifications and feature of the device contribute to the performance level of a person in order to meet the expected output in a given time and to practice ergonomically upon doing such activity.

Recommendations

Based on the findings and conclusions of the study, the following recommendations are forwarded.

Development of trolley type drawing and art table is recommended to enhance ergonomically in inclined or slanting position upon doing painting subject and to utilize some of its mechanism such as adjustable holder for faster and more effective performance while doing painting and mechanical drawing. Therefore, it may suggest that fabrication of the device will be more durable if material used will be aluminum tubular type framing for table laminated fiber panel interior and exterior of the device. Further innovation, includes using laptop for Computer-Aided Design attach to the drafting table covered with pull-up type cover of the said table.

Further study is also recommended that other material except wooden framing, fibre optics or aluminium cladding and moulding should be utilized to create this equipment. Further study is encouraged also to improve the features of the device that could be added or tested to enhance the design of the product. Other materials for locking such as wing nut or bolt and nut should be utilized to make the product strong and easier to assemble.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

The device is being develop as to its usefulness, mechanism and space saver in order to enhance creativity and accuracy which is limited only for mechanical drawing and painting.

The development of this device is to produced very good and effective result in terms of free hand drawing or painting as well as for mobility.

A device wherein drawing paper and watercolor or illustration board is involved for sketching and art application in terms of artistic manipulation and add aesthetic function to enhance mechanical and technical drawing.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

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This product relates to a develop trolley type drawing and art table particularly to mechanical drawing and painting. This device is an alternative to conventional single unit drawing table and easel stand which often used in subject area of visual techniques, architectural design and drafting.

It furthers focused on the mechanism of its parts and function, acceptability of the said device enhance the performance level of the user for project making.

The processes or services transferred will be part of the community extension program of Industrial Technology department. The project is intended to make aware of the research based develop study to help enrich capability of drafting student, visual artist for conducting develop research into a livelihood skills for those out of school youth, SMEs furniture makers, carpenters and in turn will become a part of community livelihood program.

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

Unit	Quantity	Description	Unit price	Amount
1	Sheet	1/2 thickness x 4 x 8 Marine Ply board	650	650
6	Pieces	1 x 2 x 8' wooden stick for framing, support and stand	120	720
2	Pieces	1/2 x 2" Hinges	15	30
1	Set	1/2 x 2" Piano Hinges	250	250
4	Pieces	3/4" x 2 1/2" Bolt and nut	50	200
4	Pieces	#300 Sanding Paper	25	100
1	Liter	Epoxy primer	150	150
1	Sache	Wood Glue	50	50
1/2	Kilo	1" common wire nails	60	60
2	Set	Caster wheel	150	300
		Total Amount		2,510
		2,510 (0.40) =Labor (1,004.00)		
		Summary of Expenses		3,514

End user of the technology are those in the field of Engineering and Construction Firm, Junior-Senior

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High School Vocational Technology-Drafting Major and Industrial Arts subject.

(17) **Acknowledgments** (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) **References** (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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Product design specifications http://homepages.cae.wisc.edu/~me349/lecture_notes/product_design_spec.pdf (Acc. 15 June 2016)

Development of an ergonomically designed drafting table and chair for engineering students of LPU-laguna based on anthropometric measurement; Ryan Jeffrey Curbano, CIE, MSc Lyceum of the Philippines – Laguna LPU-Laguna Journal of Multidisciplinary Research Vol. 4 No.3 September 2015

International Research Journal of Engineering and Technology (IRJET) e-ISSN: 2395-0056 Volume: 05 Issue: 05 | May-2018 www.irjet.net p-ISSN: 2395-0072 © 2018, IRJET | Impact Factor value: 6.171 | ISO 9001:2008 Certified Journal | Page 1690 Design and Fabrication of Easy Handling TrolleyShubham S. Shiwarkar1, Sagar D. Pairag2, Prof. Shailendra R. Zaveri3

S. Sainath, K. Surender, V. Vikram Arvind, “Automated Shopping Trolley for Super Market Billing System”, International Journal of Computer Applications (0975 – 8887) International Conference on Communication, Computing and Information Technology (ICCCMIT-2014)

Al Nazirul Bin Haron, “Design And Fabricate Multipurpose Trolley”, Faculty Of Mechanical Engineering, Universiti Malaysia Pahang, December 2010

Design and Development of a Multi-Purpose Trolley Article in Global Journal of Enterprise Information System · May 2017 DOI: 10.18311/gjeis/2017/15871

Multifunctional drawing board table CN216231424U • 2022-04-08 • Applicants CIGARETTE TABLE UNIV, Inventors DONG GUIHAN; Classifications IPC A47B27/02; B44D3/00; Earliest priority: 2021-03-09 • Earliest publication: 2022-04-08

Drawing Apparatus, Drawing Method, And Drawing Program Ep3098777A1Applicants Socionext Inc [Jp] Inventors Kodama Yuko [Jp] (B1) • 2016-11-30 • Socionext Inc [Jp] Earliest Priority: 2015-05-28 • Earliest Publication: 2016-11-30

Foldable painting stand US7431257B1 • 2008-10-07 • Inventors DAVIS DAVE [US]; DAVIS M SHANE [US] Earliest priority: 2005-02-10 • Earliest publication: 2008-10-07

Trolley Ep2195220A2 (B1) • 2010-06-16 • Go Trolley Gmbh; Applicants Go Caddy Promotion Packaging G [At] Inventors Pachler Robert [At] Earliest Priority: 2007-10-05 • Earliest Publication: 2009-04-16